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From Structure to Spectrum

$\text{Sr}_2\text{PrSbO}_6$ - A Wide Bandgap Perovskite Shining in UV Optoelectronics

Md. Mohiuddin, Alamgir Kabir

Department of Physics, University of Dhaka, Dhaka 1000, Bangladesh

With a stable cubic structure and wide bandgap, $\text{Sr}_2\text{PrSbO}_6$ shows strong optical activity in the UV range, highlighting its potential in UV LEDs and solar cells.

Abstract

The search for stable, efficient materials in renewable energy and optoelectronics has led to the exploration of double perovskites. In this study, we examine $\text{Sr}_2\text{PrSbO}_6$, a double perovskite known for its cubic symmetry and thermodynamic stability. Leveraging advanced computational methods.

Methodology

This study employed DFT-based methods to explore the structural, electronic, and optical properties of $\text{Sr}_2\text{PrSbO}_6$. Using VASP, we applied the PBE functional for accuracy in exchange-correlation interactions. Key calculations included band structure and density of states, followed by optical analyses. These properties were examined to determine $\text{Sr}_2\text{PrSbO}_6$'s suitability for optoelectronic applications.

Structural property

$\text{Sr}_2\text{PrSbO}_6$ forms a stable cubic perovskite structure with lattice parameters of $4.267 \text{ \AA}$. Its tolerance factor of 0.968 confirms structural robustness, .

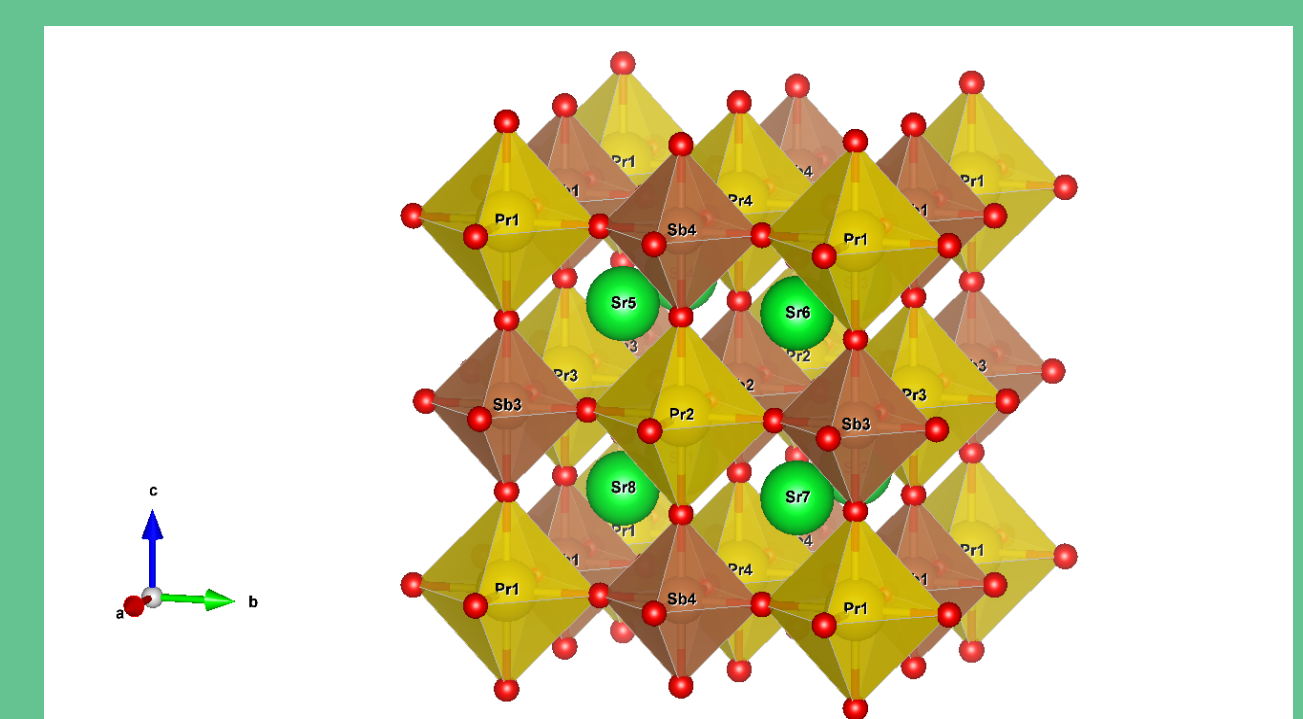


Fig: Polyhedral View of $\text{Sr}_2\text{PrSbO}_6$

Optical Property

Through DFT, we evaluated $\text{Sr}_2\text{PrSbO}_6$'s dielectric function, revealing significant optical activity in the UV range. The material shows a high refractive index and strong absorption starting from 4 eV, positioning it well for UV-visible optoelectronic applications. Additionally, its sharp energy loss peak at around 10.5 eV suggests potential for plasmonic applications.

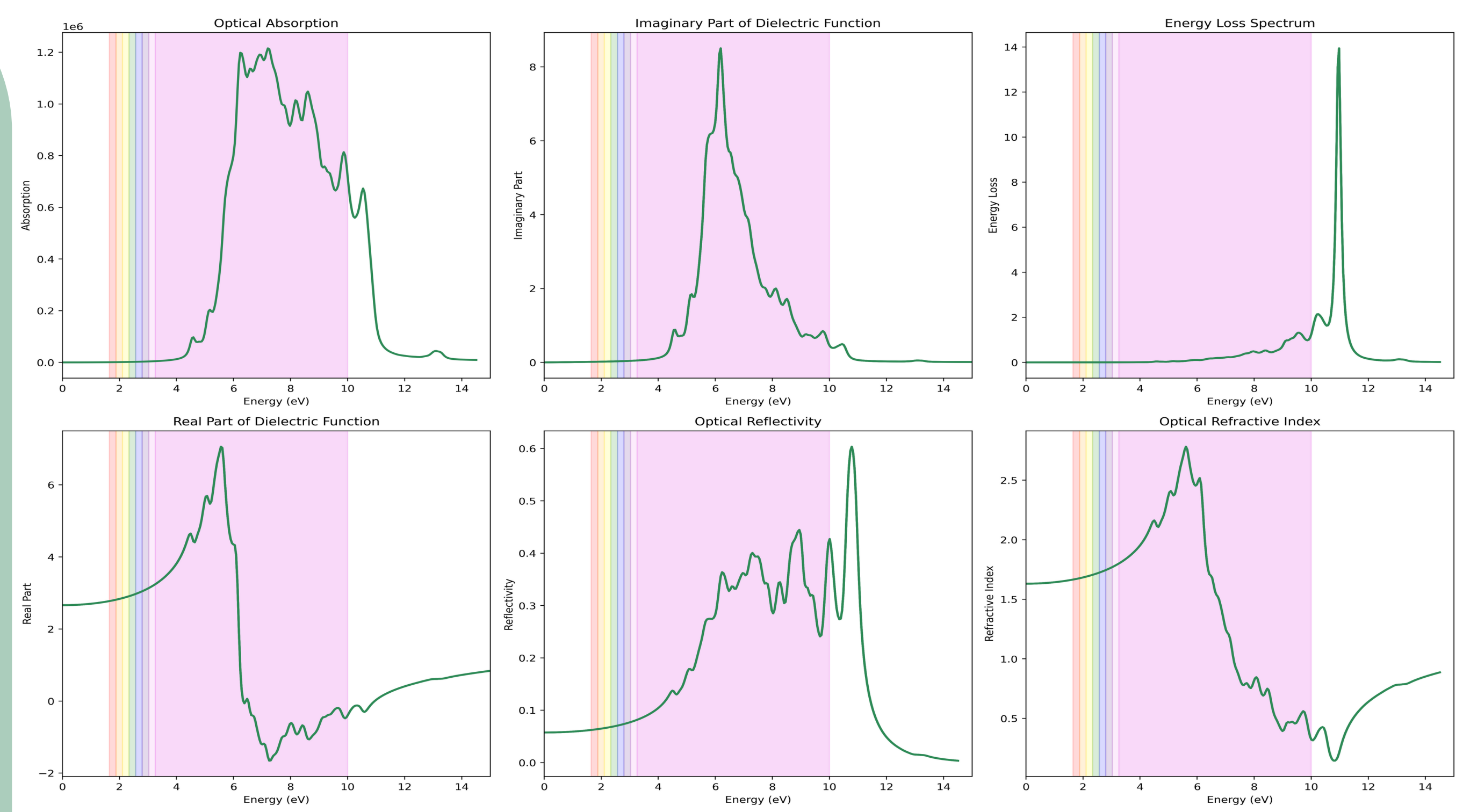


Fig: Different Optical Properties of $\text{Sr}_2\text{PrSbO}_6$

Electronic Property

DFT analysis shows that $\text{Sr}_2\text{PrSbO}_6$ has a direct bandgap of 3.488 eV, making it a wide-bandgap semiconductor ideal for UV-stable, non-magnetic optoelectronic applications. The valence band is dominated by O 2p orbitals, while Pr 5d states lead the conduction band, enhancing optical transition potential.

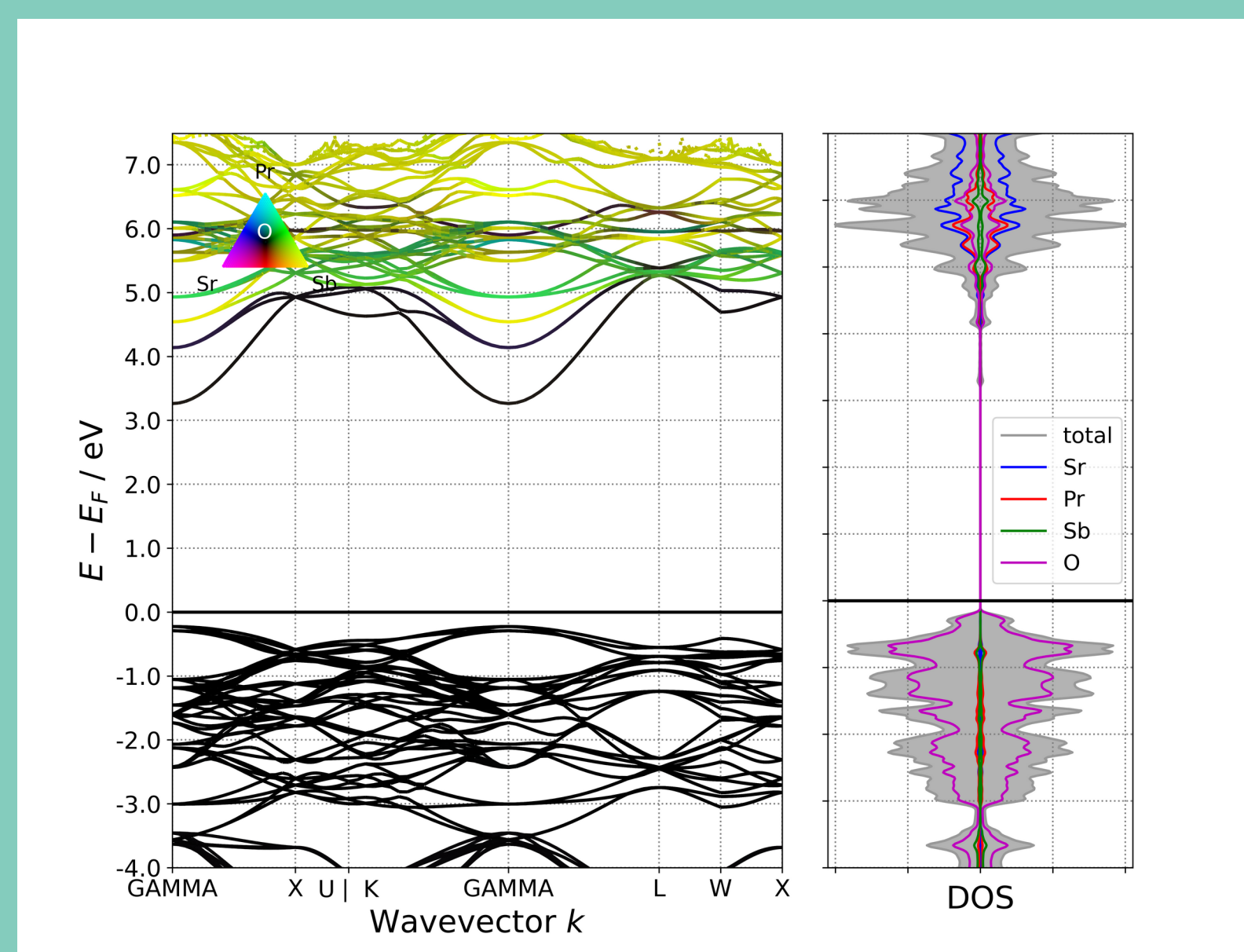


Fig: Band Structure and the Density of state of $\text{Sr}_2\text{PrSbO}_6$

Chemical Stability

$\text{Sr}_2\text{PrSbO}_6$ shows high thermodynamic stability, with a significantly negative formation energy of -22.77 eV per atom, indicating resistance to decomposition under standard conditions. Decomposition into SrO , Pr_2O_3 , and Sb_2O_3 is possible only in extreme scenarios.

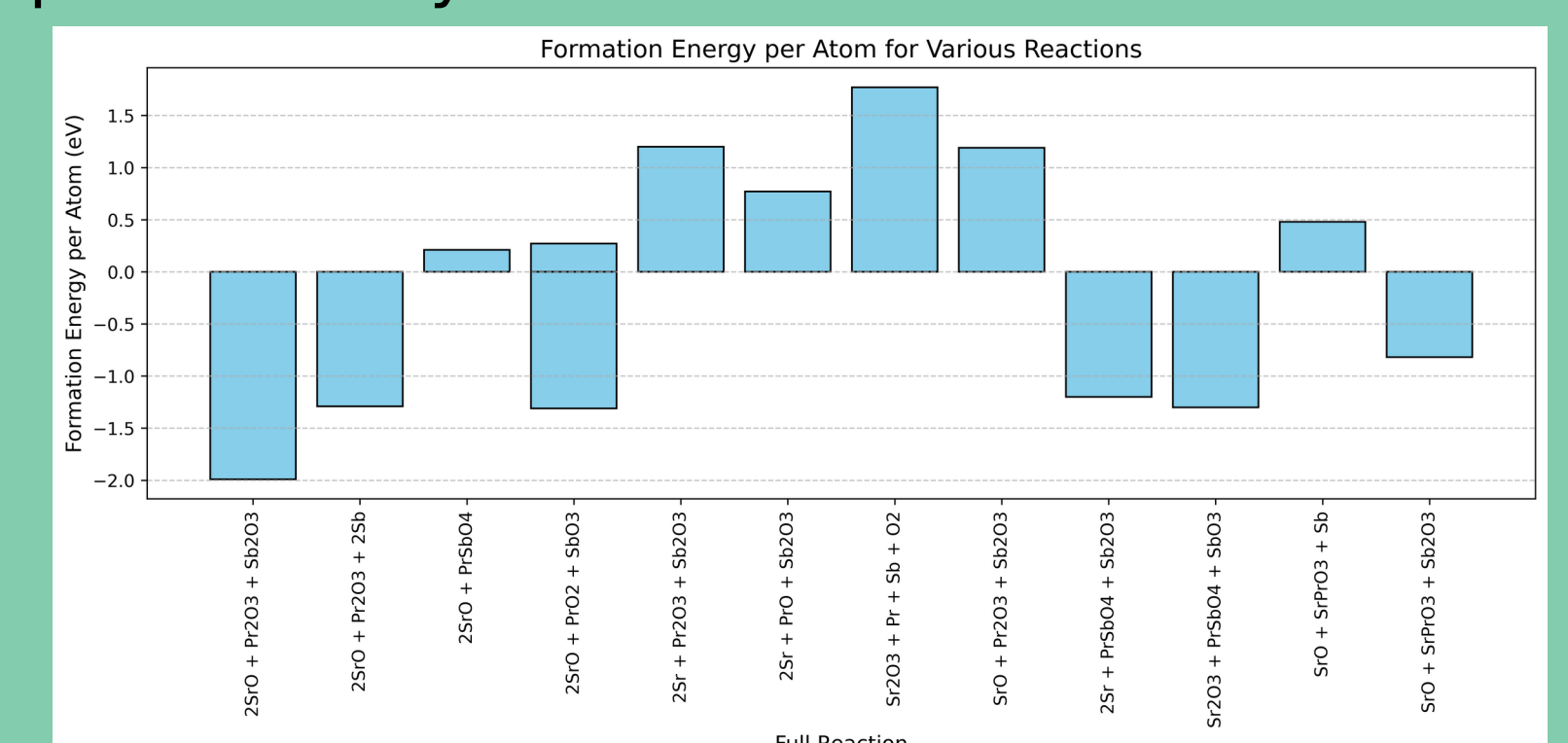


Fig: Formation energy of decomposition reaction



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